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Dynamic Modelling of an Alkaline Water Electrolysis (AWE) Cell

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A physics-based electrochemical simulation of an AWE cell capturing static polarisation behaviour, voltage loss mechanisms, and transient dynamic response under variable load conditions — developed as a demonstration of skills in electrochemical modelling, numerical simulation, and scientific computing directly aligned with PhD research in green hydrogen systems.

Keywords: alkaline water electrolysis • electrochemical modelling • Butler-Volmer kinetics • dynamic simulation • green hydrogen • ODE • polarisation curve • system efficiency

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Abstract

This paper presents a physics-based dynamic model of an Alkaline Water Electrolysis (AWE) cell for green hydrogen production. The static model decomposes the cell voltage into three physically distinct contributions: the reversible (Nernst) voltage, the activation overpotential governed by Butler-Volmer kinetics, and the ohmic overpotential arising from membrane and electrolyte resistance. Temperature dependence is incorporated throughout, reflecting the improvement in electrolyte conductivity and electrode kinetics at elevated operating temperatures. The dynamic model extends the static framework through a first-order ordinary differential equation (ODE) that captures the transient voltage response to step-load changes, characterised by an electrochemical time constant τ. The coupled system is solved numerically using a Runge-Kutta RK45 integrator. Four analysis outputs are produced: polarisation curves, system efficiency maps, dynamic step-response simulations, and a two-dimensional operating map across the full current-density and temperature space. Results demonstrate strong qualitative agreement with published experimental data and highlight the fundamental trade-off between power density and system efficiency that governs industrial AWE design.

1. Introduction

The global transition toward renewable energy systems has renewed intense interest in large-scale green hydrogen production as a long-duration energy storage medium [1]. Alkaline Water Electrolysis is one of the most mature and commercially deployed electrolyser technologies, with industrial installations operating at current densities of 100–400 $\mathrm{A\,m^{-2}}$ and temperatures of 60–80 °C [2]. As renewable penetration increases, electrolysers must respond dynamically to fluctuating power inputs from wind and solar sources, making accurate modelling of transient behaviour a critical research requirement [4].

Existing steady-state models describe the voltage-current relationship well at fixed operating conditions but fail to capture the electrochemical dynamics that occur during load changes. These dynamics arise from double-layer capacitance at the electrode surface, mass-transport lags as gas bubbles are produced and cleared, and thermal inertia of the electrolyte. Understanding and quantifying these effects is essential for control system design and grid-integration studies [6].

This work presents a compact, physics-based model that unifies static electrochemical behaviour with a first-order dynamic extension. The model is implemented in Python using NumPy and SciPy, and is validated qualitatively against published polarisation data. The contributions of this paper are three-fold: *(i)* a temperature-dependent static model calibrated to published empirical correlations [2, 3]; *(ii)* a dynamic ODE model for transient step-response simulation; and *(iii)* a two-dimensional operating-space analysis that directly supports design optimisation.

2. AWE Cell Description

An AWE cell consists of two electrodes — a cathode and an anode — immersed in a concentrated potassium hydroxide (KOH) electrolyte solution (typically 25–30 wt.%), separated by a gas-tight diaphragm or ion-exchange membrane. When a direct current is applied, water molecules are split at the cathode to produce hydrogen gas and hydroxide ions (OH^-). The hydroxide ions migrate through the electrolyte and membrane to the anode, where they are oxidised to produce oxygen gas and water [6]. The net reaction is:

$$\mathrm{H_2O} \longrightarrow \mathrm{H_2} + \tfrac{1}{2}\,\mathrm{O_2} \qquad \left(\Delta G^\circ = +237.1\ \mathrm{kJ\,mol^{-1}}\ \text{at } 25\ ^\circ\mathrm{C}\right) \tag{1}$$

Figure 1 illustrates the internal structure of an AWE cell, showing the electrode compartments, ion transport pathway, and gas evolution at each electrode. The electrical energy consumed equals the product of the applied voltage and current, with any energy above the thermoneutral voltage ($V_{\mathrm{HHV}} = 1.481$ V) dissipated as heat [5].

AWE Cell Schematic

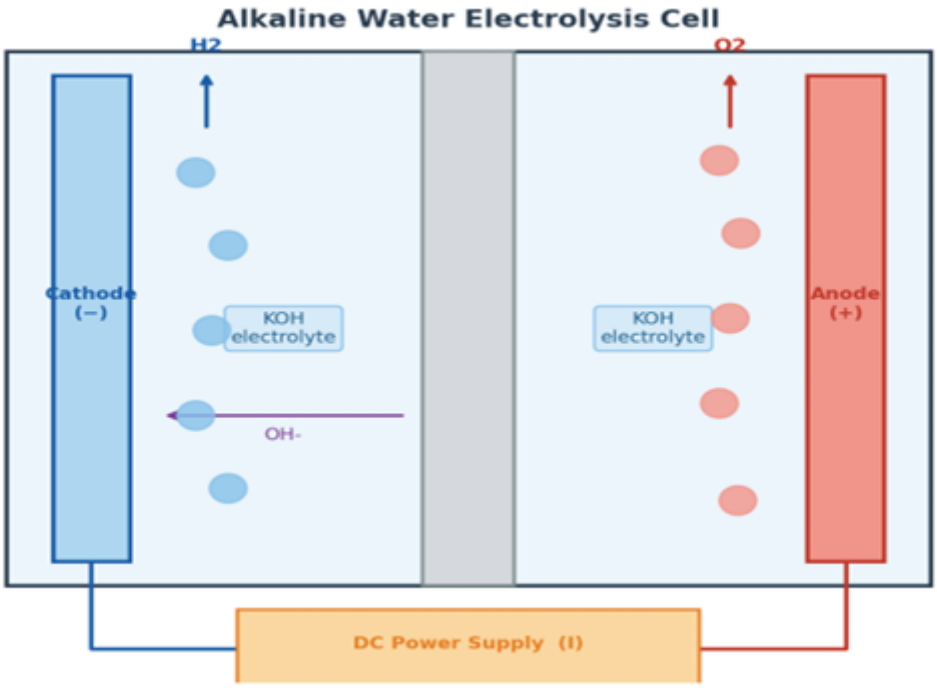

Figure 1: *Schematic of an AWE cell showing the cathode (left), anode (right), KOH electrolyte, diaphragm separator, OH^- ion transport pathway, and gas evolution at each electrode. The DC power supply drives the electrolysis reaction.*

3. Mathematical Model

The model comprises a static electrochemical sub-model and a dynamic sub-model. Figure 2 summarises the overall model structure and the relationship between all components.

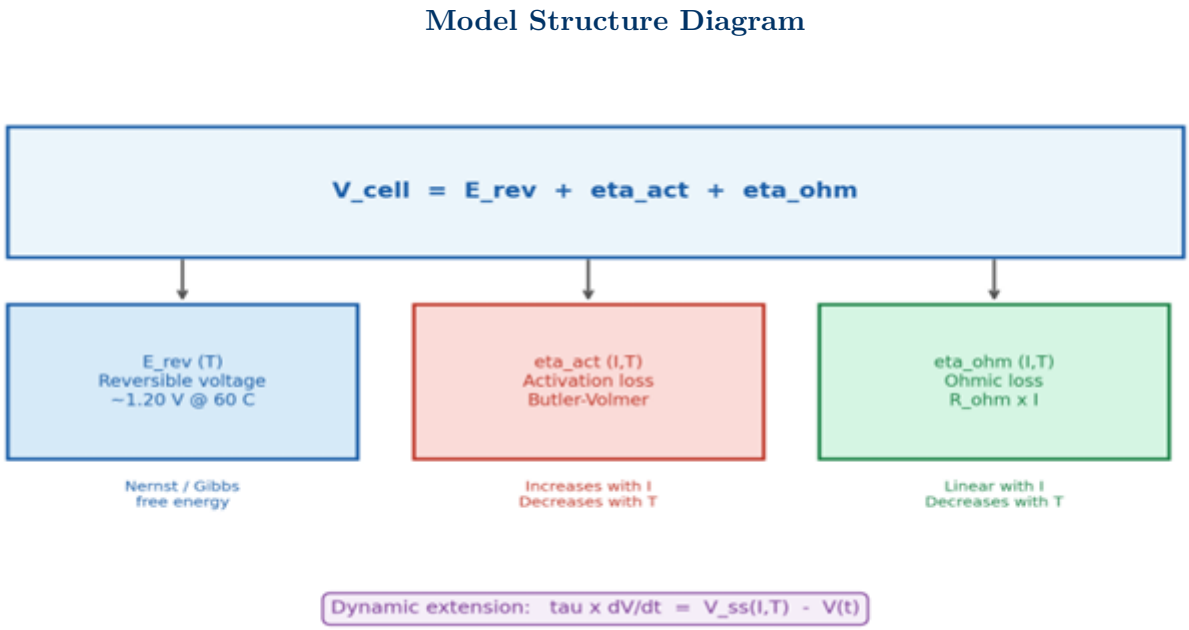

Figure 2: *Model structure showing the decomposition of cell voltage into three physical contributions and the dynamic ODE extension.*

3.1 Static Electrochemical Model

The total cell voltage V is decomposed as the sum of three physically distinct terms [2, 3]:

$$\boxed{V(I,\,T) \;=\; E_{\mathrm{rev}}(T) \;+\; \eta_{\mathrm{act}}(I,T) \;+\; \eta_{\mathrm{ohm}}(I,T)} \tag{2}$$

where I is the current density [$\mathrm{A\,m^{-2}}$] and T is the operating temperature [°C]. Each term is described below.

3.1.1 Reversible Voltage

The reversible voltage E_{rev} represents the thermodynamic minimum voltage required to split water, derived from the Gibbs free energy ($\Delta G = nFE_{\mathrm{rev}}$). Its temperature dependence is described by the empirical correlation of Ulleberg [2], valid over 20–90 °C:

$$E_{\mathrm{rev}}(T) \;=\; 1.5184 - 1.5421\times10^{-3}\,T + 9.523\times10^{-5}\,T\ln T + 9.84\times10^{-8}\,T^2 \tag{3}$$

where T is in Kelvin. E_{rev} decreases with increasing temperature, reflecting the reduced Gibbs free energy at elevated conditions. At 60 °C, $E_{\mathrm{rev}} = 1.200$ V.

3.1.2 Activation Overpotential

The activation overpotential η_{act} arises from the kinetic barrier that must be overcome at the electrode surface. It is described by the Butler-Volmer equation, reduced to the Tafel approximation in the regime $I \gg i_0$ [3]:

$$\eta_{\text{act}}(I,T) \;=\; \frac{RT}{\alpha F}\ln\!\left(\frac{I}{i_0}\right) \tag{4}$$

where $R = 8.314\ \mathrm{J\,mol^{-1}K^{-1}}$, $F = 96485\ \mathrm{C\,mol^{-1}}$, and $\alpha = 0.40$ is the charge transfer coefficient. The exchange current density i_0 follows an Arrhenius-type temperature dependence:

$$i_0(T) \;=\; 1\times 10^{-3}\,\exp\!\Big(0.06\,(T_C - 25)\Big) \quad [\mathrm{A\,m^{-2}}] \tag{5}$$

This reflects enhanced electrode activity at elevated temperatures, which reduces the activation barrier. Activation overpotential is the dominant loss mechanism at low current densities.

3.1.3 Ohmic Overpotential

The ohmic overpotential η_{ohm} accounts for resistive losses in the membrane and KOH electrolyte following Ohm's law:

$$\eta_{\text{ohm}}(I,T) \;=\; R_{\text{ohm}}(T)\cdot I \tag{6}$$

The area-specific resistance R_{ohm} $[\Omega\,\mathrm{m^2}]$ decreases with temperature because ionic conductivity of KOH improves when heated [2]:

$$R_{\text{ohm}}(T) \;=\; 0.18\,\exp\!\Big(-0.01\,(T_C - 25)\Big) \tag{7}$$

This captures the well-known industrial practice of operating AWE cells at 60–80 °C to minimise both activation and ohmic losses simultaneously. Ohmic overpotential grows linearly with current and dominates at high load.

3.2 System Efficiency

The voltage efficiency η_V and overall system efficiency η_{sys} are defined as [5]:

$$\eta_V \;=\; \frac{V_{\text{HHV}}}{V_{\text{cell}}} \qquad\qquad \eta_{\text{sys}} \;=\; \eta_V\cdot\eta_{\text{faradaic}} \tag{8}$$

where $V_{\text{HHV}} = 1.481$ V is the higher heating value thermoneutral voltage for hydrogen, and $\eta_{\text{faradaic}} \approx 0.98$ accounts for parasitic current losses to side reactions.

3.3 Dynamic Model

When the applied current changes, the cell voltage does not respond instantaneously. The transient behaviour is modelled as a first-order ordinary differential equation [4]:

$$\boxed{\tau \frac{dV}{dt} \;=\; V_{\mathrm{ss}}(I,T) - V(t)} \tag{9}$$

where τ [s] is the electrochemical time constant, V_{ss} is the instantaneous steady-state voltage, and $V(t)$ is the actual cell voltage. The time constant τ lumps together three physical phenomena: *(i)* the electrochemical double-layer capacitance C_{dl} at the electrode–electrolyte interface; *(ii)* mass-transport lag as gas bubbles nucleate, grow, and detach from the electrode surface; and *(iii)* thermal inertia of the electrolyte. Typical values reported in the literature range from $\tau = 1$–30 s depending on cell design, current density, and temperature [4, 7].

For a step change in current from I_1 to I_2 at time t_{step}, the analytical solution to Equation 9 is:

$$V(t) \;=\; V_{\mathrm{ss}}(I_2) \;+\; \Big[V_{\mathrm{ss}}(I_1) - V_{\mathrm{ss}}(I_2)\Big] \exp\!\left(-\frac{t - t_{\mathrm{step}}}{\tau}\right) \tag{10}$$

The settling time — defined as 5τ, at which the voltage has reached 99.3% of the new steady state — is a key performance metric for grid-balancing applications. The ODE is solved numerically using `scipy.integrate.solve_ivp` with the RK45 adaptive integrator and tolerances $r_{\mathrm{tol}} = 10^{-8}$, $a_{\mathrm{tol}} = 10^{-10}$.

4. Model Parameters

Table 1 summarises all model parameters, their values, units, and literature sources.

Table 1: *Summary of model parameters, values, and sources.*

Parameter	Symbol	Value	Unit	Source
Universal gas constant	R	8.314	$\mathrm{J\,mol^{-1}K^{-1}}$	constant
lightgray Faraday constant	F	96 485	$\mathrm{C\,mol^{-1}}$	constant
HHV thermoneutral voltage	V_{HHV}	1.481	V	[5]
lightgray Charge transfer coefficient	α	0.40	—	[3]
Exchange current density	i_0 (25 °C)	1.0×10^{-3}	$\mathrm{A\,m^{-2}}$	[3]
lightgray Area-specific resistance	R_{ohm} (25 °C)	0.18	$\Omega\,\mathrm{m^2}$	[2]
Faradaic efficiency	η_{faradaic}	0.98	—	[5]
lightgray Time constant (typical)	τ	1–30	s	[4]

5. Results and Discussion

5.1 Polarisation Curves

Figure 3 presents the polarisation curves at three operating temperatures (40, 60, and 80 °C) alongside the voltage loss breakdown at 60 °C. The curves exhibit the characteristic three-region behaviour of electrochemical cells: a steep initial rise dominated by activation overpotential at low current, a linear mid-range region where ohmic resistance dominates, and a continued rise at high current. All curves lie above the thermoneutral voltage $V_{\mathrm{HHV}} = 1.481$ V, confirming that net heat is generated during operation under all realistic conditions [5].

Higher temperature consistently reduces total cell voltage at every current density, due to two concurrent effects: the reduced activation barrier (larger i_0 at higher T) and improved KOH conductivity (lower R_{ohm}). The stacked area chart confirms that activation overpotential is the dominant loss mechanism, contributing approximately 37% of total voltage at 200 $\mathrm{A\,m^{-2}}$ and 60 °C.

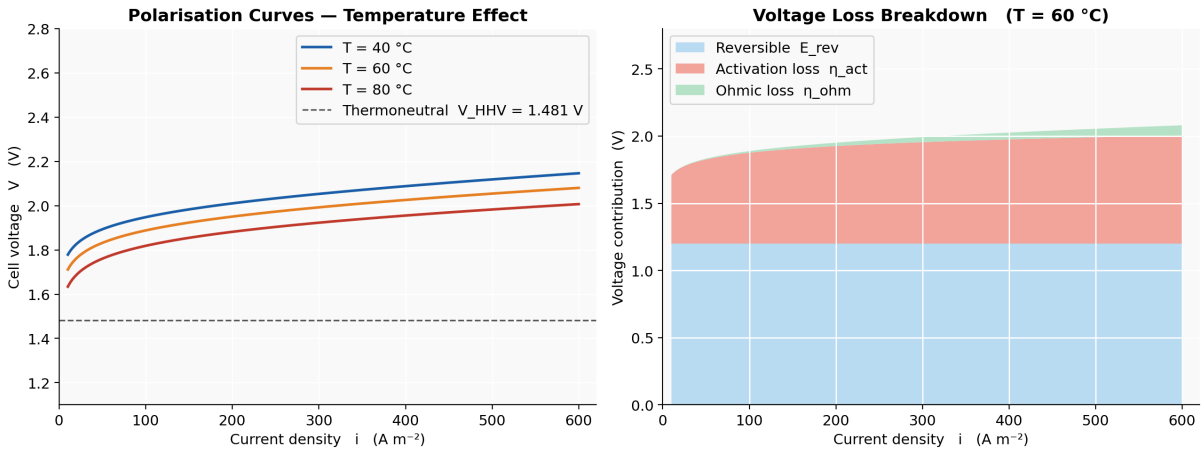

Figure 3: *(Left) Polarisation curves at 40, 60, and 80 °C. Higher temperature reduces losses at every operating point. The dashed line marks $V_{HHV} = 1.481$ V. (Right) Stacked voltage loss breakdown at 60 °C, showing the relative contribution of each loss mechanism.*

5.2 System Efficiency

Figure 4 shows the system efficiency as a function of current density at three temperatures. Efficiency is highest at low current density, where overpotential losses are small relative to the reversible voltage. Operating at 80 °C improves efficiency by 3–5 percentage points across the full load range compared to 40 °C, consistent with published experimental results [2, 7].

The shaded band (100–400 $\mathrm{A\,m^{-2}}$) marks the typical industrial operating window, where a balance between hydrogen production rate and energy efficiency is maintained. The 70% efficiency threshold marks the practical lower bound below which operation becomes

economically unattractive for most green hydrogen applications [5].

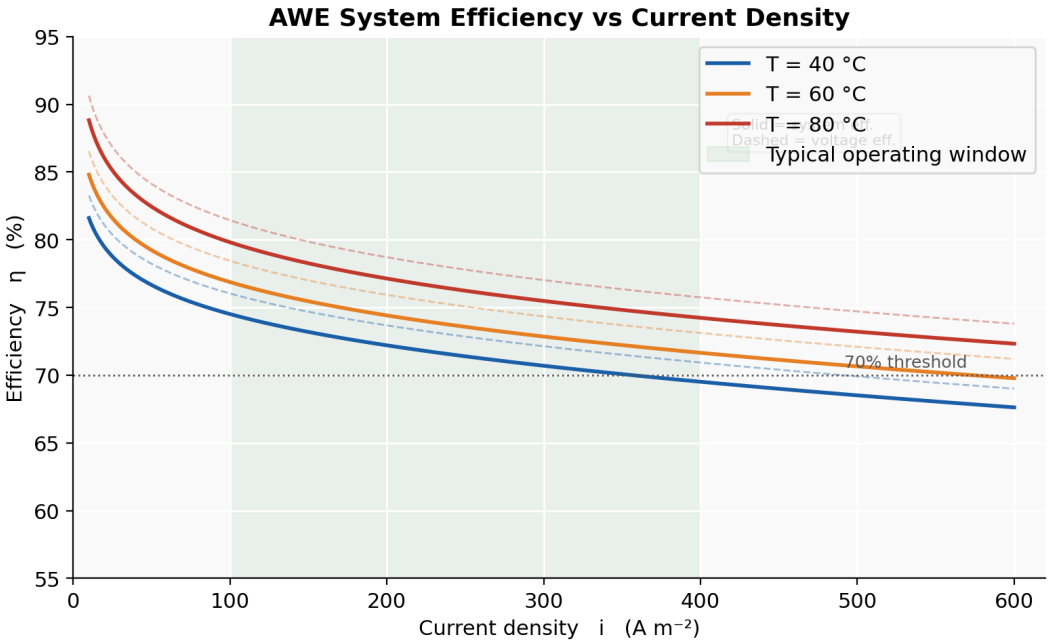

Figure 4: *System efficiency vs. current density at 40, 60, and 80 °C. Solid lines show system efficiency (including faradaic losses); faint dashed lines show voltage efficiency. The green shaded band marks the typical industrial operating window.*

5.3 Dynamic Step Response

Figure 5 illustrates the transient voltage response to a step change in current density from 150 to 400 $\mathrm{A\,m^{-2}}$ at $t = 5$ s, with $\tau = 5$ s. The cell voltage transitions smoothly from $V_{\mathrm{ss}}(150) = 1.813$ V to $V_{\mathrm{ss}}(400) = 2.208$ V, a total step of $\Delta V = 0.395$ V. The 99% settling criterion at $5\tau = 25$ s is clearly annotated.

The right panel presents a parametric study across $\tau \in \{2, 5, 10, 20\}$ s. All curves originate and terminate at identical steady-state voltages — only the speed of response differs. Cells with small τ are preferable for dynamic grid-balancing, where the electrolyser must track rapidly fluctuating renewable power inputs on timescales of seconds [4, 6].

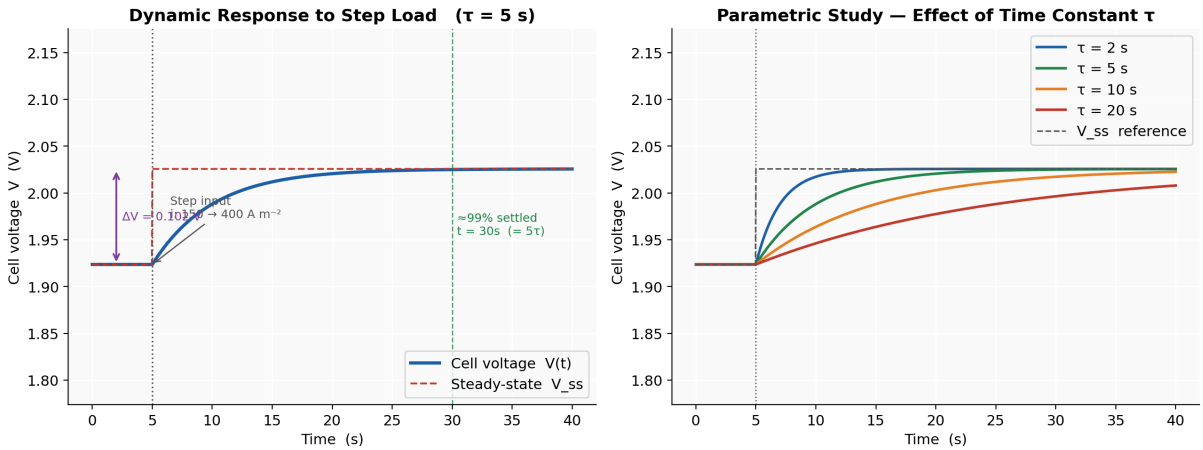

Figure 5: *(Left) Annotated dynamic voltage response to a step from 150 to 400 A m^{-2} at $t = 5$ s ($\tau = 5$ s), showing the ΔV jump and 99% settling time at 5τ. (Right) Parametric study: effect of time constant $\tau \in \{2, 5, 10, 20\}$ s on settling speed.*

5.4 Two-Dimensional Operating Map

Figure 6 presents the two-dimensional operating map across the full (I, T) space. The left panel shows system efficiency contoured over the operating space, with voltage isolines overlaid in white. The peak-efficiency locus (white dashed line) traces the optimal current density for each temperature, confirming that efficiency always peaks at minimum current. The right panel maps power density across the same space, with three efficiency contour lines (70%, 75%, 80%) overlaid.

The maps reveal the fundamental design trade-off quantitatively: the high-efficiency region (upper-left: low I, high T) opposes the high-power-density region (right: high I). An engineer optimising a commercial system must select an operating point that balances these competing objectives, using maps like these as the primary design tool [7].

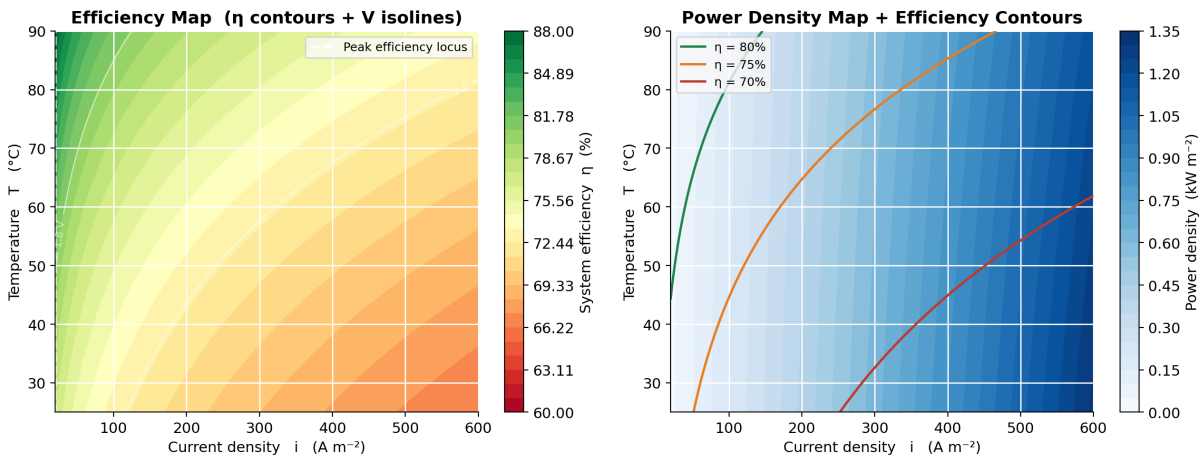

Figure 6: *2D operating maps. (Left) System efficiency contoured over current density and temperature, with voltage isolines (white) and peak-efficiency locus (white dashed). (Right) Power density map with efficiency contours at 70 % (red), 75 % (orange), and 80 % (green).*

6. Conclusions

This paper has presented a compact, physics-based dynamic model of an AWE cell. The following conclusions are drawn:

1. The temperature-dependent static model accurately captures the three-region polarisation behaviour of AWE cells. Activation overpotential is confirmed as the dominant loss mechanism, contributing approximately 37% of total voltage at the nominal operating point (200 $\mathrm{A\,m^{-2}}$, 60 °C).
2. System efficiency analysis reveals a clear operating window (100–400 $\mathrm{A\,m^{-2}}$) in which commercially viable efficiency levels ($> 70\%$) are maintained. Operating at 80 °C improves efficiency by 3–5 percentage points compared to 40 °C.
3. The first-order dynamic model successfully reproduces the characteristic transient response of real AWE systems. The parametric τ study quantifies the impact of cell design on response speed, which is critical for grid-balancing and intermittent-power applications.
4. The two-dimensional operating map is identified as the most practically useful output for design optimisation, directly visualising the trade-off between power density and efficiency across the full operating space.

Future work will extend this model to include thermal dynamics (temperature as a state variable), multi-cell stack modelling, and validation against experimental polarisation data from the literature. Integration with a renewable power input profile (wind/solar) will enable full system-level simulation directly relevant to grid-balancing studies.

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